

Is Your LLM Outdated? A Deep Look at Temporal Generalization

Chenghao Zhu^{1,2}, Nuo Chen¹, Yufei Gao¹,

Yunyi Zhang¹, Prayag Tiwari³, Benyou Wang^{1,2*}

¹ The Chinese University of Hong Kong, Shenzhen

² Shenzhen Research Institute of Big Data

³ Halmstad University

wangbenyou@cuhk.edu.cn

Abstract

The rapid advancement of Large Language Models (LLMs) has led to the development of benchmarks that consider temporal dynamics, however, there remains a gap in understanding how well these models can generalize across temporal contexts due to the inherent dynamic nature of language and information. This paper introduces the concept of temporal generalization in LLMs, including bias in past and future generalizations. Then we introduce FreshBench, a new evaluation framework that employs fresh text and event prediction for assessing LLMs’ temporal adaptability, ensuring the evaluation process free from data leakage and subjective bias. The experiment shows significant temporal biases and a decline in performance over time. Our findings reveal that powerful models, while initially superior, tend to decline more rapidly in future generalization. Additionally, powerful open-source models demonstrate better long-term adaptability compared to their closed-source counterparts. Our code is available at <https://github.com/FreedomIntelligence/FreshBench>

1 Introduction

Proprietary large language models (LLMs) OpenAI (2023); GoogleAI (2023); Anthropic (2023); Anil et al. (2023) has been paralleled by efforts in the open-source community to democratize LLMs. Meanwhile, the rapid advancement of LLM democratization Touvron et al. (2023) emphasizes the necessity for dynamic and robust benchmarks that accurately reflect their evolving capabilities. Traditional benchmarks, broadly classified into knowledge-based assessments and open-dialogue evaluations, each present unique challenges. Knowledge-based assessments, exemplified by MMLU Hendrycks et al. (2021) and C-Eval Huang et al. (2023), are prone to data manipulation, raising concerns about their real-world

applicability and the risk of data leakage Wei et al. (2023); Golchin and Surdeanu (2024). Open-dialogue evaluations like MT-Bench Zheng et al. (2023) and Alpaca-Eval Li et al. (2023a), rely on subjective human or model-based judgments, making their outcomes susceptible to the influence of dialogue structure, and thereby, potentially compromising assessment validity.

Given the limitations of traditional methods, researchers are increasingly developing new approaches to evaluate models in diverse and dynamic contexts. For instance, recent efforts have introduced methods that assess models’ abilities to handle temporal evolution, contextual understanding, and real-time information updates Fatemi et al. (2024); Kasai et al. (2024); Vu et al. (2023); Chen et al. (2021); Zhang and Choi (2021); Liška et al. (2022). While methods like RealTimeQA and FreshQA focus on continuously updating knowledge for real-time processing Kasai et al. (2024); Vu et al. (2023), others like LatestEval Li et al. (2023b) emphasize recent text-based comprehension. In contrast, our work shifts the focus to evaluating models’ world understanding through prediction, using a fact-based approach that minimizes bias and ensures objectivity in assessment. This method enhances robustness against data contamination and focuses on the model’s capacity for accurate, real-world event prediction.

We believe that the evaluation of LLMs should be conducted in scenarios that are resistant to hacking manipulation, with results that are objective and **free from evaluator biases** Chen et al. (2024). Additionally, time as a dynamic dimension, can also ensure robust and relevant evaluations. This paper explores an alternative way to **evaluate large language models**, i.e., **on temporal generalization** where time is considered as an important aspect. This approach significantly reduces the issue of data contamination by limiting access to future content. Moreover, a retrospective perspective on

*Benyou is the corresponding author.

time could be beneficial for temporal bias analysis for LLMs.

To achieve this goal, we design two scenarios. The first is to have the model generate text based on recent data such as arXiv papers, news articles, new books, and Wikipedia content. An excellent model should be able to generalize these new texts with high language likelihood without data leakage. The second scenario focuses on future event prediction. An excellent model should be able to understand the current context, integrate world knowledge, and effectively generalize to future scenarios. In these two scenarios, temporal generalization evaluation has two main advantages: eliminating the possibility of data leakage through temporal boundaries and enabling objective evaluation results without relying on subjective interpretations.

Our analysis indicates that using validation loss of new text to evaluate models is mostly effective. However, we observe that while performance on language likelihood is a strong indicator of a model’s linguistic capabilities, it does not always align with specific abilities measured by existing benchmarks, which may represent discrete knowledge or other capacities that are important but not directly related to language modeling.

Our **contributions** are as follows: 1) We define and quantify temporal generalization and bias, establishing a basis for understanding and assessing the capabilities of LLMs over time. 2) We proposed a benchmark Freshbench to test the capabilities of LLMs in predicting future events. This benchmark is dynamically updated to reflect the latest data, ensuring that our evaluations remain relevant and accurate in the face of rapidly evolving data environments. It experimentally assesses the performance of existing LLMs across various time-frames, providing insights into their historical and ongoing performance stability and adaptability.

2 Conceptualization of Temporal Generalization

In this section, we will first conceptualize *Temporal Generalization*, then *Temporal Biases* and *Temporal Degeneration* in Sec. 2.1. We will discuss ways to measure /textitTemporal Biases and Degeneration in 2.2.

2.1 Temporal Generalization

LLMs face the ongoing challenge of remaining applicable as new text evolves. Consequently, it

is crucial for these models to not only understand and generate text based on seen (e.g., past) data but also to anticipate and adapt to future data. This inspires temporal generalization as below.

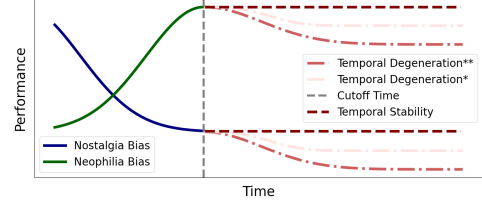


Figure 1: Schematic figure of temporal generalization. In general, prior to the cutoff time, an LLM might exhibit a *Nostalgia* bias that is skewed toward past dates (i.e., aligning better with historical data) or a *Neophilia* bias that is skewed toward future dates (favoring more recent information). However, after the cutoff time, the model’s understanding of future trends declines, leading to temporal degeneration.

Definition 2.1. Temporal Generalization refers to the LLMs’ ability to align with the contexts of the past, present, and future.

This requires the models to integrate and apply knowledge from historical and current data to adapt to the evolution of language and emerging trends. Temporal generalization highlights the challenge of designing models that remain effective and accurate over time, without the need for frequent re-training as new data is introduced. LLMs must maintain a delicate balance between leveraging historical data to understand long-standing language patterns and adapting to new expressions, terminology, and topics as they emerge.

2.1.1 For Models: Define Past, Present and Future

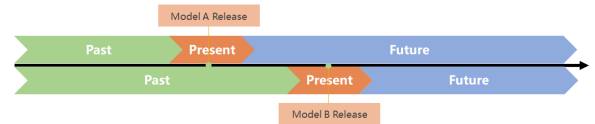


Figure 2: Schematic Diagram of 3 Periods of Models

To distinguish the *past* and the *future*, we define the cutoff time as the release date of the model. For instance, if an LLM is released in January 2024, then all events or information before January 2024 are considered part of the *past*, while everything after January 2024 is considered the *future*. Although the data after the cutoff time is not strictly future data, this approach ensures no data leakage for the model. As is shown in Figure 2, the period near the cutoff time can be referred to as the

present. We evaluate the model’s prediction accuracy across these time periods, denoted as acc_{past} , acc_{future} , and $acc_{present}$, respectively.

2.1.2 Define Temporal Bias and Temporal Degeneration

With the three time periods defined, we can define Temporal Bias and Temporal Degeneration for models.

Definition 2.2. Temporal Bias In this paper, we define temporal bias as the preferences of LLMs over time.¹

There exist two biases, namely **Nostalgia Bias** and **Neophilia Bias**, which reflect two contrasting tendencies in how these models might predict or generate text about the past, present, or future. A visual representation of these biases is provided in Fig. 1, where the trends associated with **Nostalgia** and **Neophilia** biases are plotted over time.

Nostalgia Bias refers to the over-reliance on historical data when generating text. This bias causes models to reflect perspectives, styles, or information prevalent in past training data, leading to outputs that may not account for recent developments. Essentially, **Nostalgia Bias** can make a model seem ‘stuck in the past’, limiting its ability to accurately represent the present or predict the future.

Neophilia Bias implies a model’s overemphasis on novelty and future trends, potentially at the expense of established information. This bias could result in predictions that are overly speculative and not well-grounded in historical data or current realities. Though less common due to LLM training on historical datasets, **Neophilia Bias** could be perceived from prompt structuring or interpretation of outputs that prioritize futuristic concepts over historical accuracy.

Definition 2.3. Temporal Degeneration refers to the decline in a model’s performance in the future, that is, after it is released.

In conclusion, the illustration for the above concepts is in Figure 1.

¹Temporal bias are multifaceted in previous literature. For instance, can refer to the assumption of an incorrect sequence of events that misleads our reasoning about causality [Dorsey \(2021\)](#) or user preferences over time in point-of-interest recommendations [Rahmani et al. \(2022\)](#). Recently, biases have raised more challenges in pre-trained language models [Yogaraajan et al. \(2023\)](#), particularly focusing on under-represented societies. [Sarkar and Vafa \(2024\)](#) examines lookahead bias in pre-trained language models, providing insights into how these biases affect model performance.

2.2 Assessing Temporal Generalization

Based on definition in Sec. 2.1, we turn to assess temporal generalization through the lens of 1) I: compression intelligence in Sec. 2.2.1 and II): future event prediction in Sec. 2.2.2.

2.2.1 Case I: Compression Intelligence

In the assessment of compression intelligence, metrics such as perplexity (PPL) and bits per character (BPC) offer insights into model performance. We refine our focus to BPC for a more normalized comparison. Unlike traditional BPC which is specific to character-level language models, our **adapted Bits Per Character (BPC)** metric offers a comparison between models by normalizing the log likelihood based on character count, irrespective of the tokenization method, allowing for a more **equitable assessment of model performance** across different languages and tokenization schemes, the BPC is calculated as follows:

$$BPC(T) = \frac{-\sum_{i=1}^N \log p(w_i | w_1, \dots, w_{i-1})}{\text{len-utf-8}(T)} \quad (1)$$

Here, T represents the text being analyzed, N is the number of tokens in T (which may vary depending on the tokenizer), and $\text{len-utf-8}(T)$ is the length of T when encoded in UTF-8, measured in characters. The tokens w_i correspond to segments of text which may vary depending on the tokenizer used. This variance is due to different tokenization methods splitting the text into tokens at different granularities, ranging from subword units to characters. In this paper, we investigate the compression intelligence proposed in [Huang et al. \(2024\)](#) in a temporal manner,

2.2.2 Case II: Future Event Prediction

LLMs could perform *future event prediction*, as seen in [Bonde \(2022\)](#) which employed GPT-3 in a few-shot setting to tackle binary questions from the Metaculus platform. Further, [Zou et al. \(2022\)](#) enriched the discourse by introducing a comprehensive dataset that includes forecasting questions alongside a news corpus. This approach enhances model training by incorporating varied and relevant data contexts. Moreover, [Halawi et al. \(2024\)](#) showcased how integrating news content can significantly improve the accuracy of predictions in real-time scenarios.

Building on this foundation, our research seeks to explore the temporal generalization capabilities